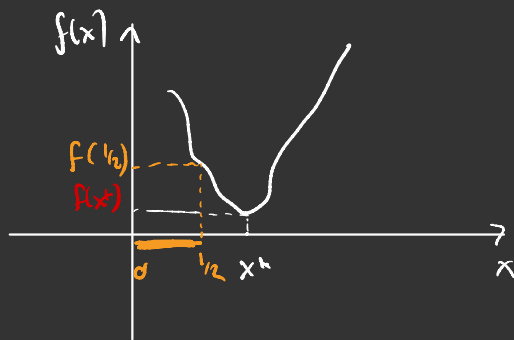


$f(x^*) = \min_{x \in [0,1]} f(x)$ → Objective

$x^k = \operatorname{argmin}_{x \in [0,1]} f(x)$



$f(x^*) = \min_{x \in [0,1], x \leq 1/2} f(x)$

$\hat{x} \approx \operatorname{argmin}_{[0, 1/2]} f(x)$

$$SS(\underline{c}) = \frac{1}{n} \sum_{i=1}^n (Y_i - c_0 - c_1 X_{i1} - \dots - c_k X_{ik})^2$$

$$\hat{\underline{\beta}} = \arg \min_{\underline{c}} SS(\underline{c}) \quad \Rightarrow \quad SS(\hat{\underline{\beta}}) = \min_{\underline{c}} SS(\underline{c})$$

$$\begin{aligned} SS(\hat{\underline{\beta}}) &= \frac{1}{n} \sum_{i=1}^n \underbrace{(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{i1} - \dots - \hat{\beta}_k X_{ik})}_{\hat{\varepsilon}_i}^2 \\ &= \frac{1}{n} \sum_{i=1}^n \hat{\varepsilon}_i^2 \end{aligned}$$

$$SSE = \sum_{i=1}^n \hat{\varepsilon}_i^2$$

$$\begin{aligned}
 E^t(\text{sat} \mid \text{hsize} = h, \text{female} = 0, \text{black} = 0) &= \beta_0 + \beta_1 h + \beta_2 h^2 \\
 E^t(\text{sat} \mid \text{hsize} = h, \text{female} = 0, \text{black} = 1) &= \beta_0 + \beta_1 h + \beta_2 h^2 + \beta_4
 \end{aligned}$$

$$= -\beta_4$$

↓ estimated by

$$-\hat{\beta}_4$$

$$SE(-\hat{\beta}_4) = SE(\hat{\beta}_4) = 15.423$$

$$t_{\text{calc}} = \frac{169.813}{15.423}$$

$$A: \text{hsize} = 3, \quad \text{female} = 1, \quad \text{black} = 0$$

$$B: \text{hsize} = 4, \quad \text{female} = 1, \quad \text{black} = 0$$

$$\text{sat}_A = E^*(\text{sat} | A) = \beta_0 + 3\beta_1 + 9\beta_2 + \beta_3$$

$$\text{sat}_B = E^*(\text{sat} | B) = \beta_0 + 4\beta_1 + 16\beta_2 + \beta_3$$

$$\Delta \text{sat} = \text{sat}_B - \text{sat}_A = \beta_1 + 7\beta_2$$

↓ estimate

$$\widehat{\Delta \text{sat}} = \hat{\beta}_1 + 7\hat{\beta}_2 \longrightarrow$$

$$\sqrt{\text{Var}(\hat{\beta}_1 + 7\hat{\beta}_2)} = \sqrt{\text{Var}(\hat{\beta}_1) + 49\text{Var}(\hat{\beta}_2) + 14\text{Cov}(\hat{\beta}_1, \hat{\beta}_2)}$$

↓

$$SE(\widehat{\Delta \text{sat}}) = SE(\hat{\beta}_1 + 7\hat{\beta}_2) = \sqrt{\widehat{\text{Var}}(\hat{\beta}_1 + 7\hat{\beta}_2)} = \sqrt{\widehat{\text{Var}}(\hat{\beta}_1) + 49\widehat{\text{Var}}(\hat{\beta}_2) + 14\widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2)}$$

We can also use

$$SE(\hat{\beta}_1 + 7\hat{\beta}_2) = SE(\underline{\lambda}^T \hat{\underline{\beta}}) = \sqrt{\underline{\lambda}^T \hat{\underline{\Omega}} \underline{\lambda}}$$

where $\underline{\lambda}^T = (0, 1, 7, 0, 0, 0)$

$$CI = [\hat{\beta}_1 + 7\hat{\beta}_2 \pm z_{0.05/2} SE(\hat{\beta}_1 + 7\hat{\beta}_2)]$$

↓

$$z_{0.025} = 1.96$$

A parameter θ , and an estimator $\hat{\theta}$

$E(\hat{\theta}) = \theta \Rightarrow \hat{\theta}$ is unbiased (definition)

$\hat{\theta} \xrightarrow{P} \theta \Rightarrow \hat{\theta}$ is consistent (definition)